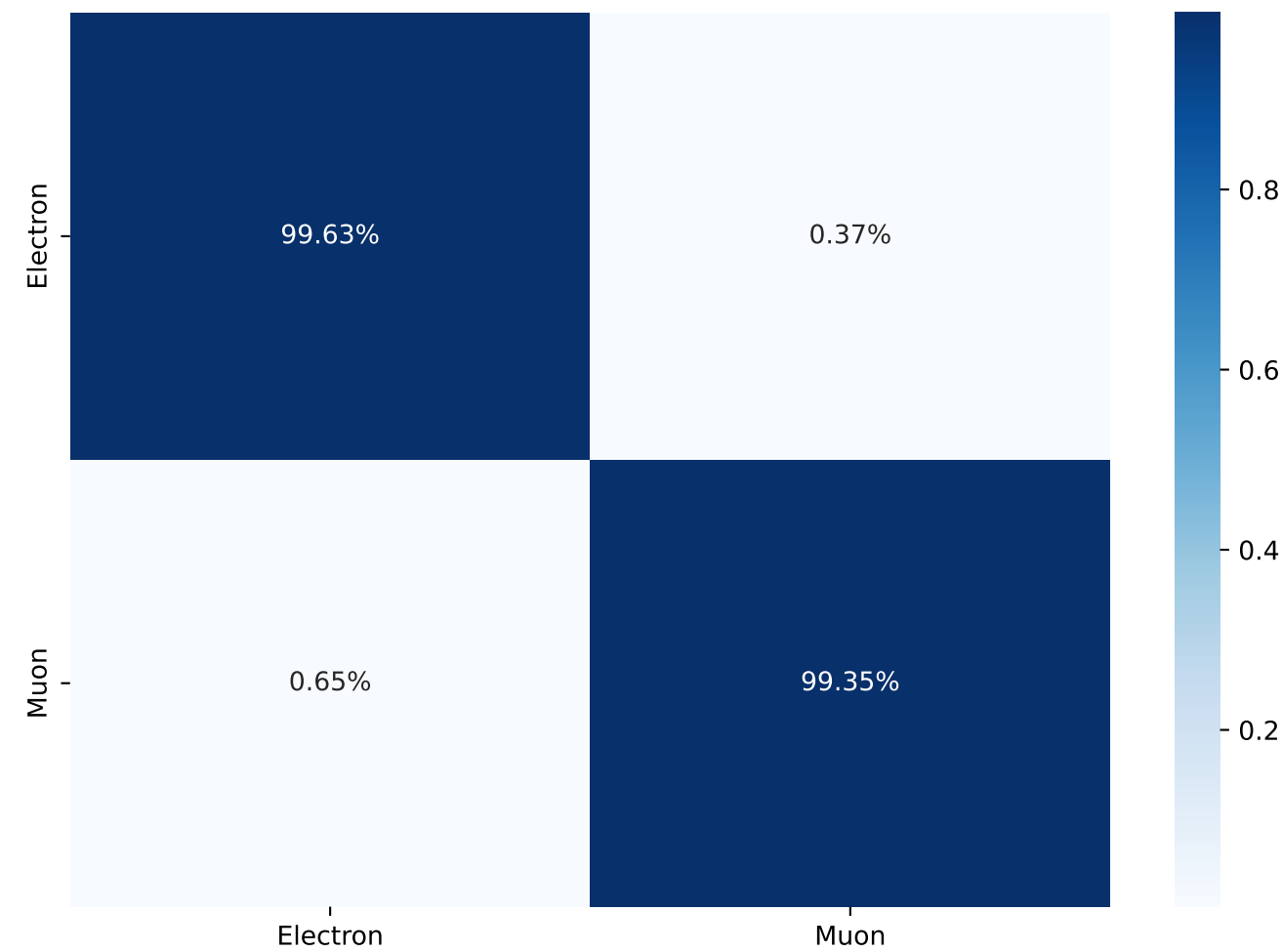
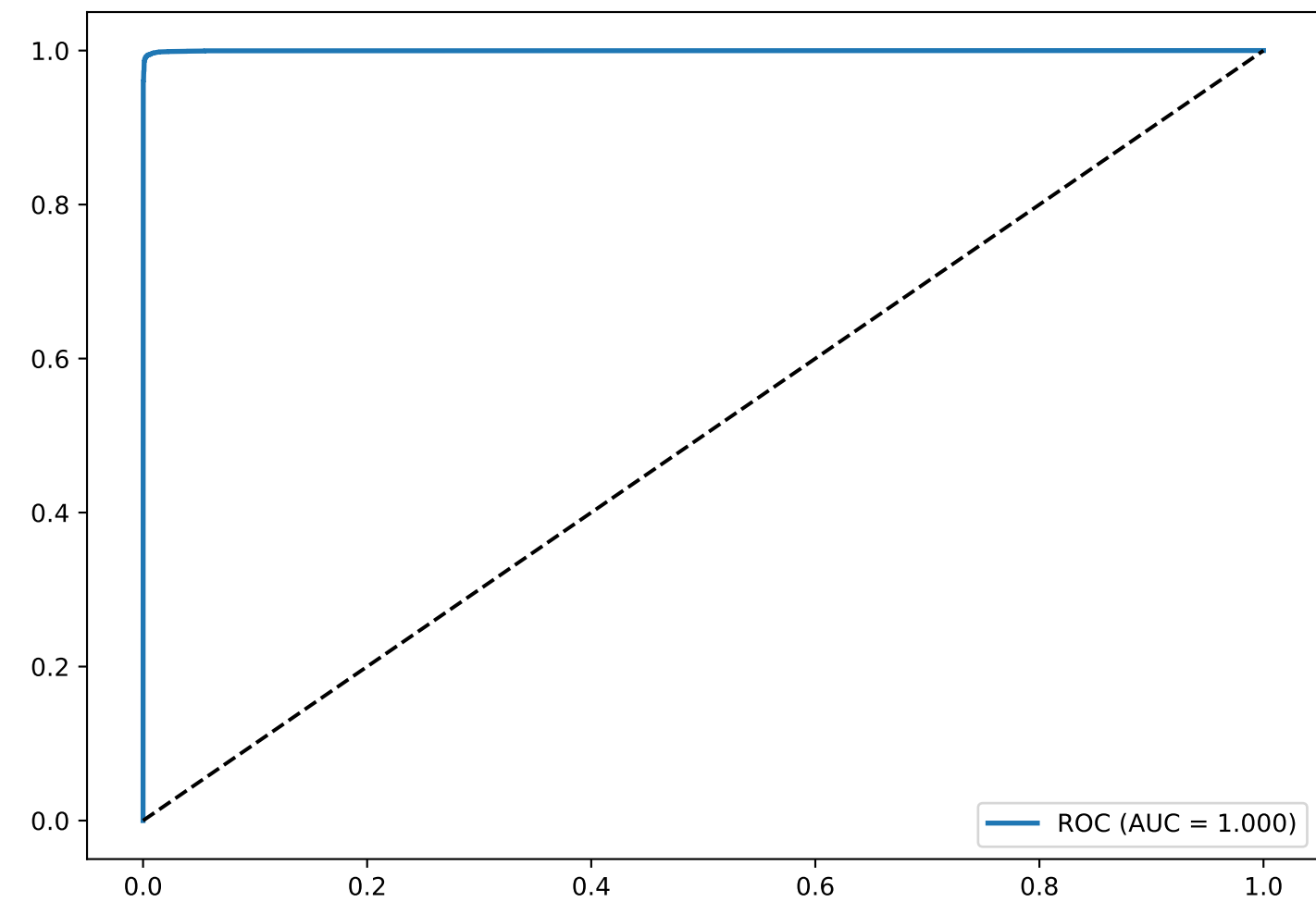


EtaRange = FullRange (TEST DATA) - Accuracy: 0.995

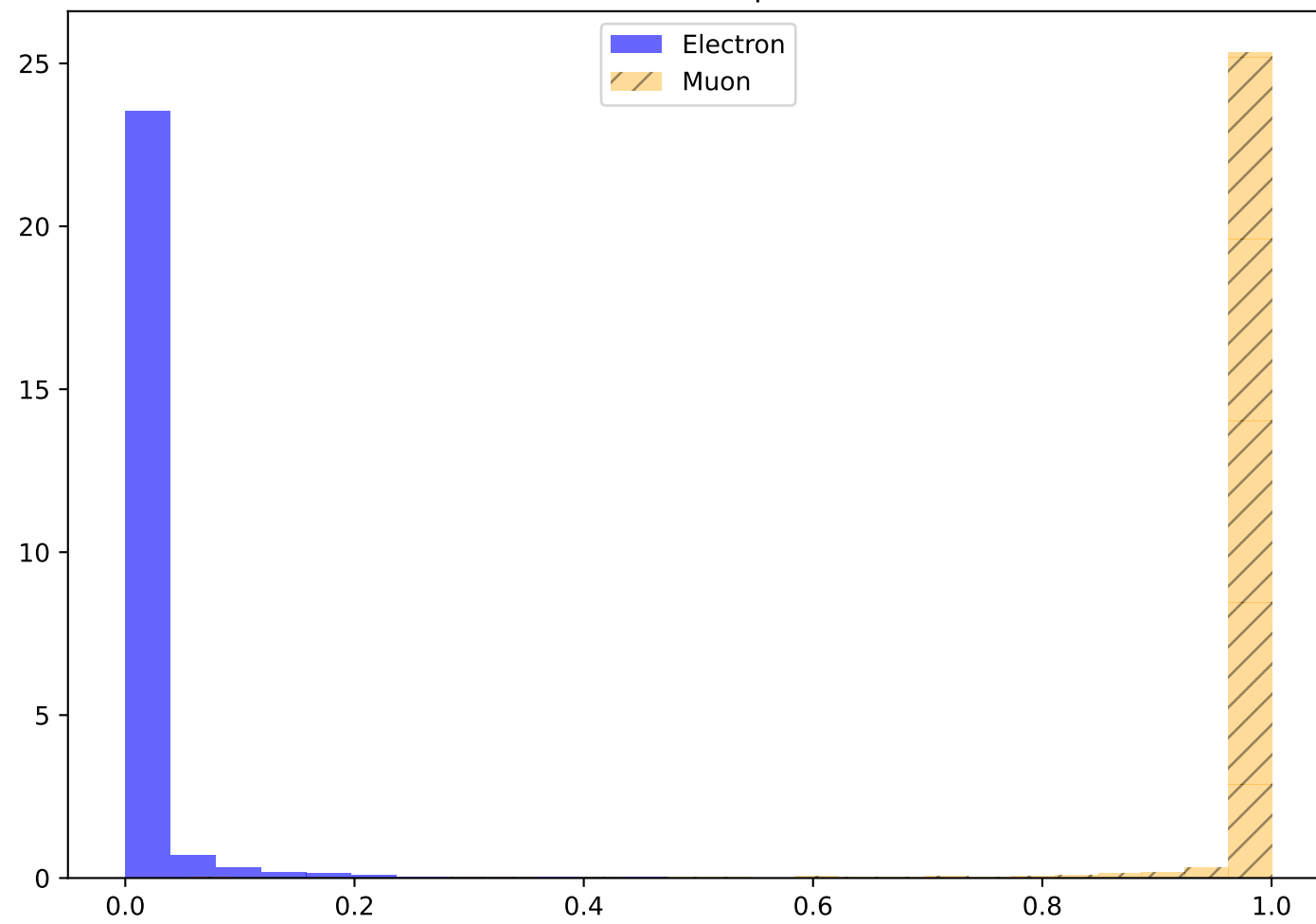
Confusion Matrix



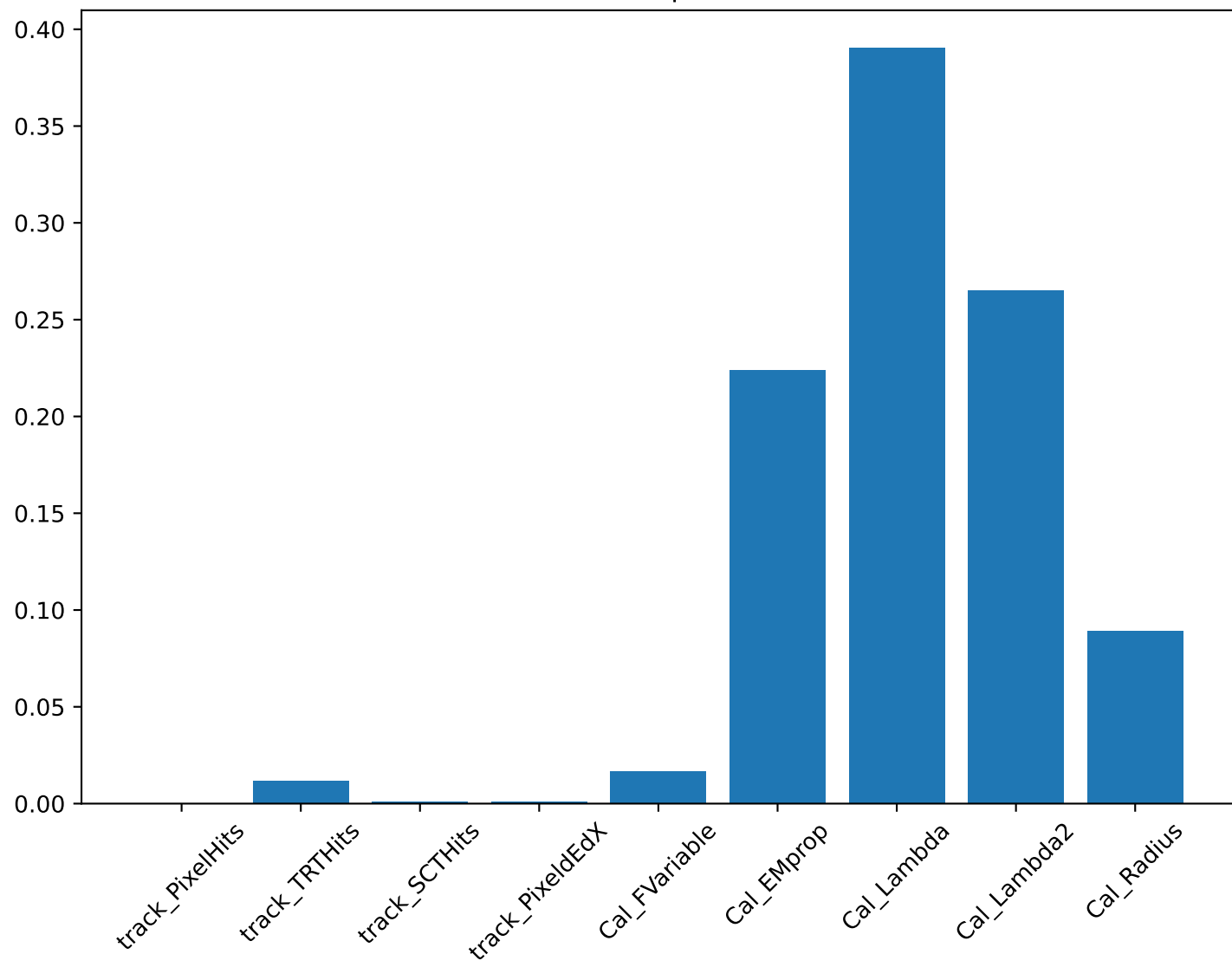
ROC Curve



Classifier Response

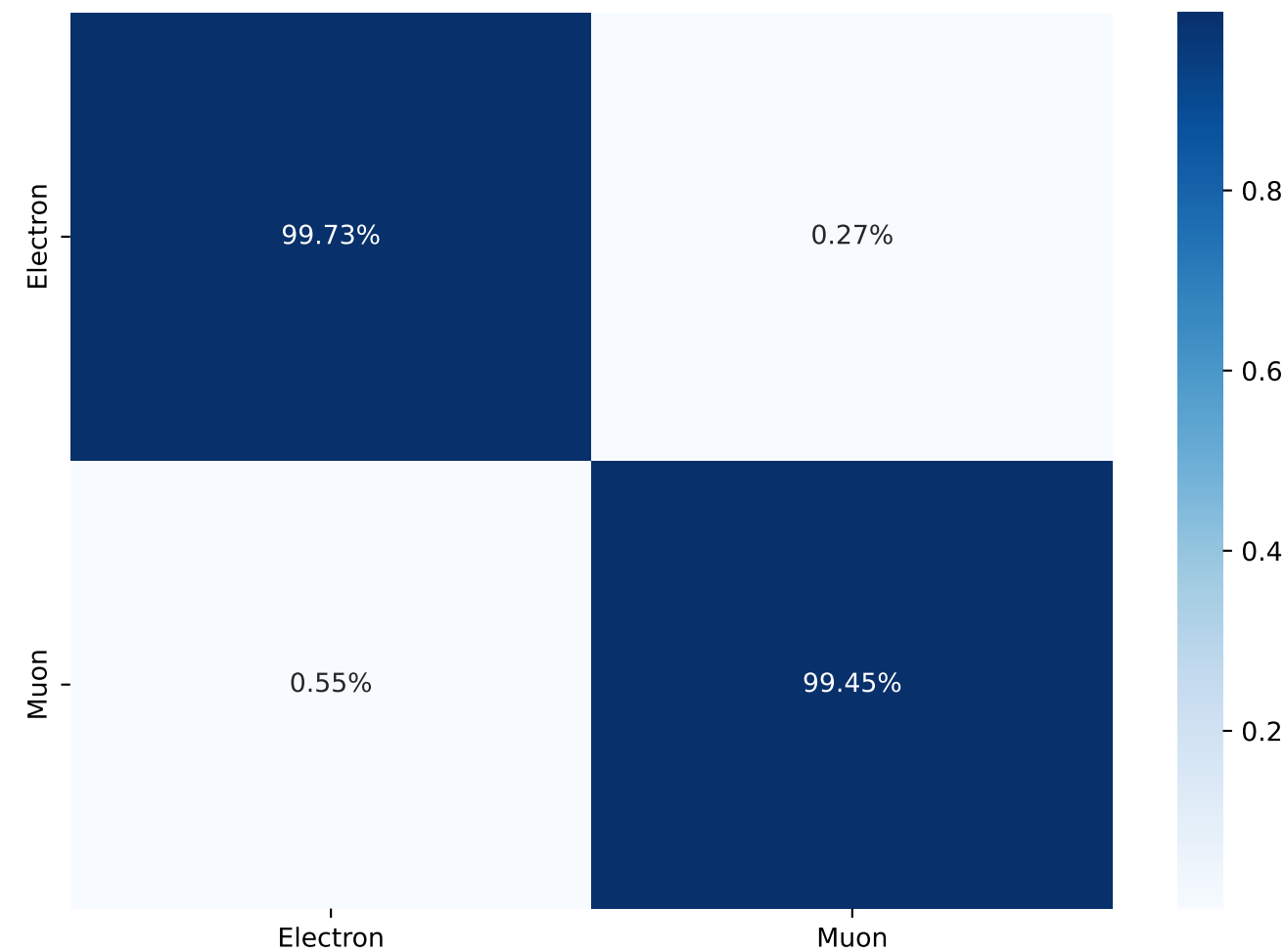


Feature Importance

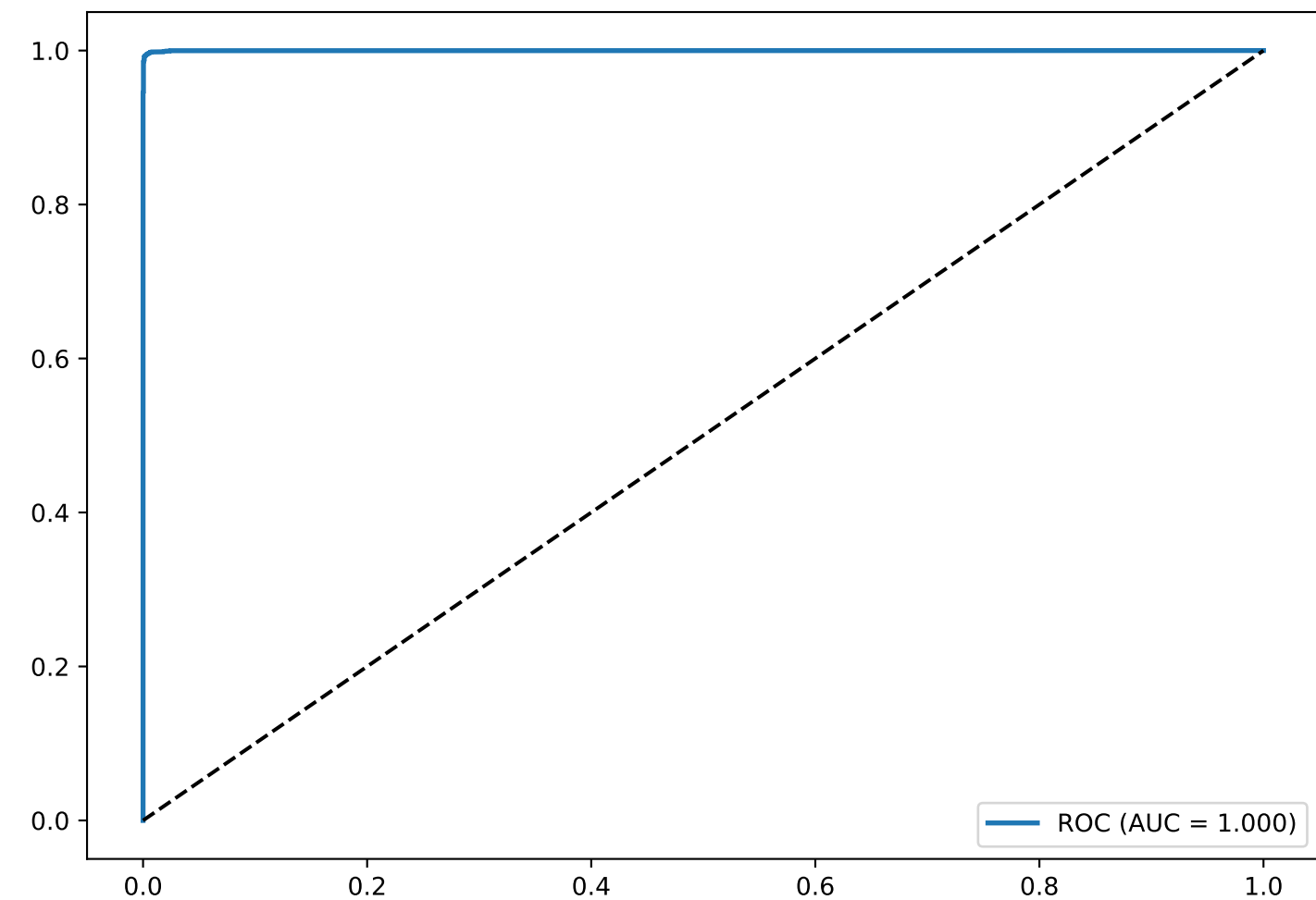


EtaRange = <1.0 (TEST DATA) - Accuracy: 0.996

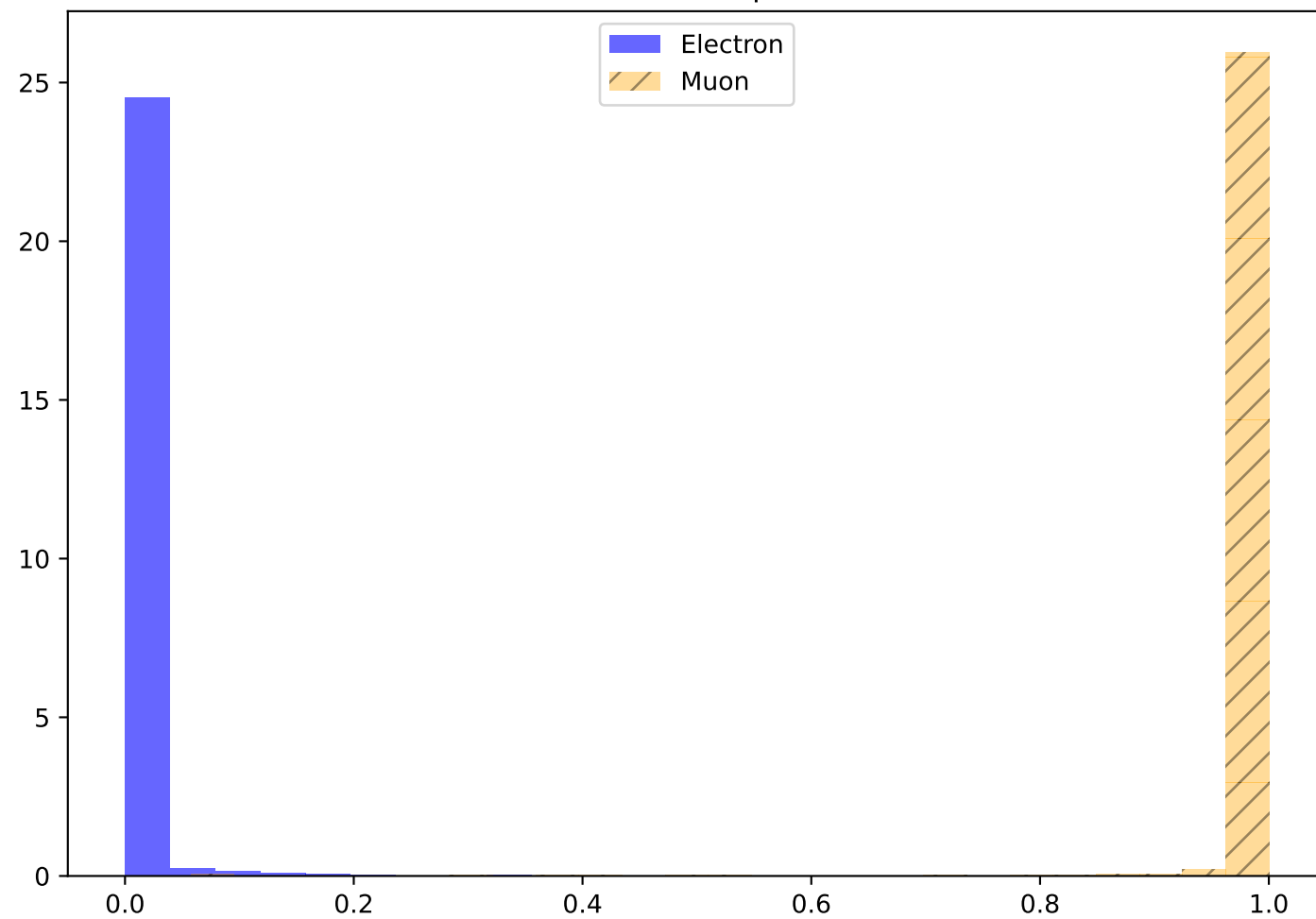
Confusion Matrix



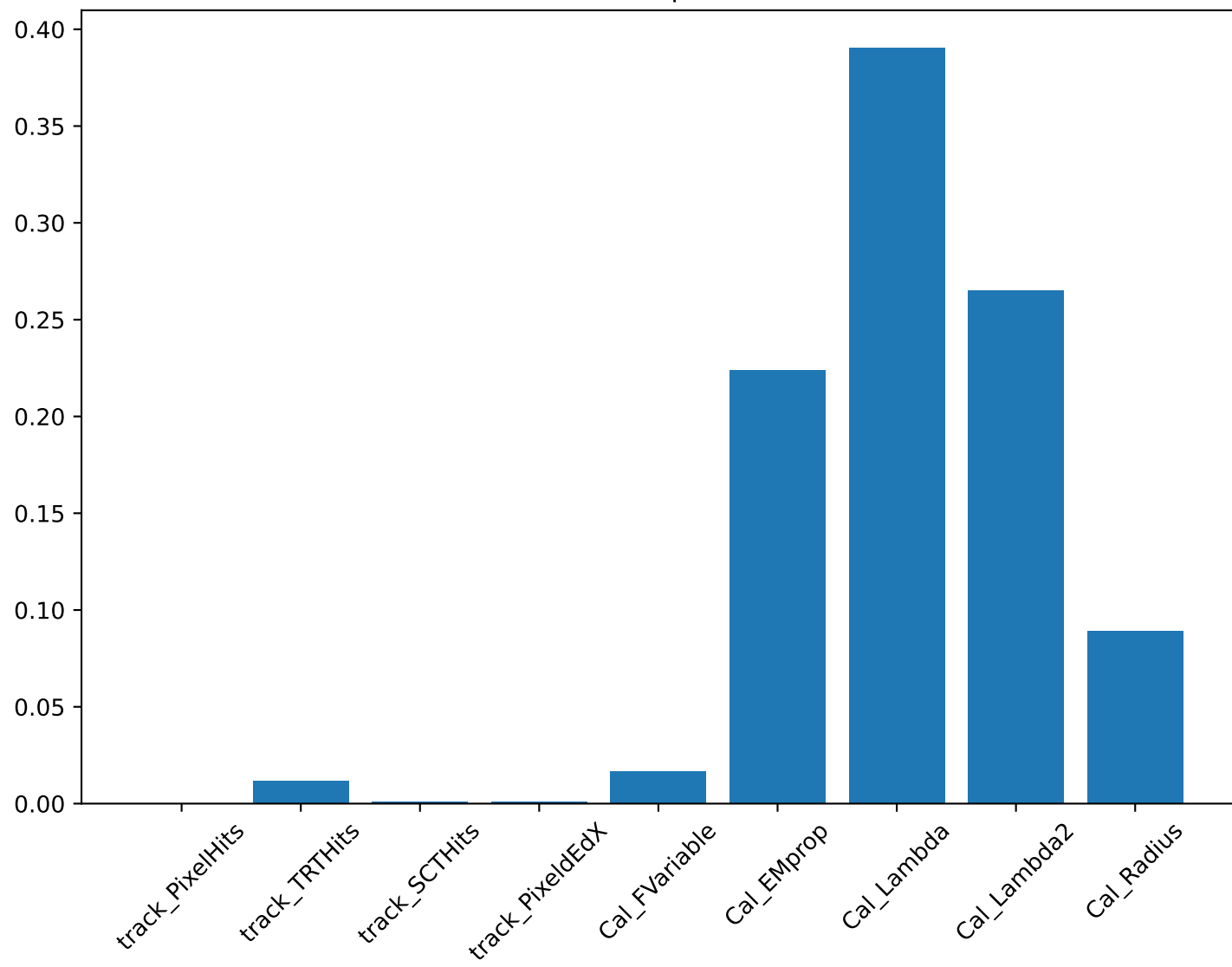
ROC Curve



Classifier Response

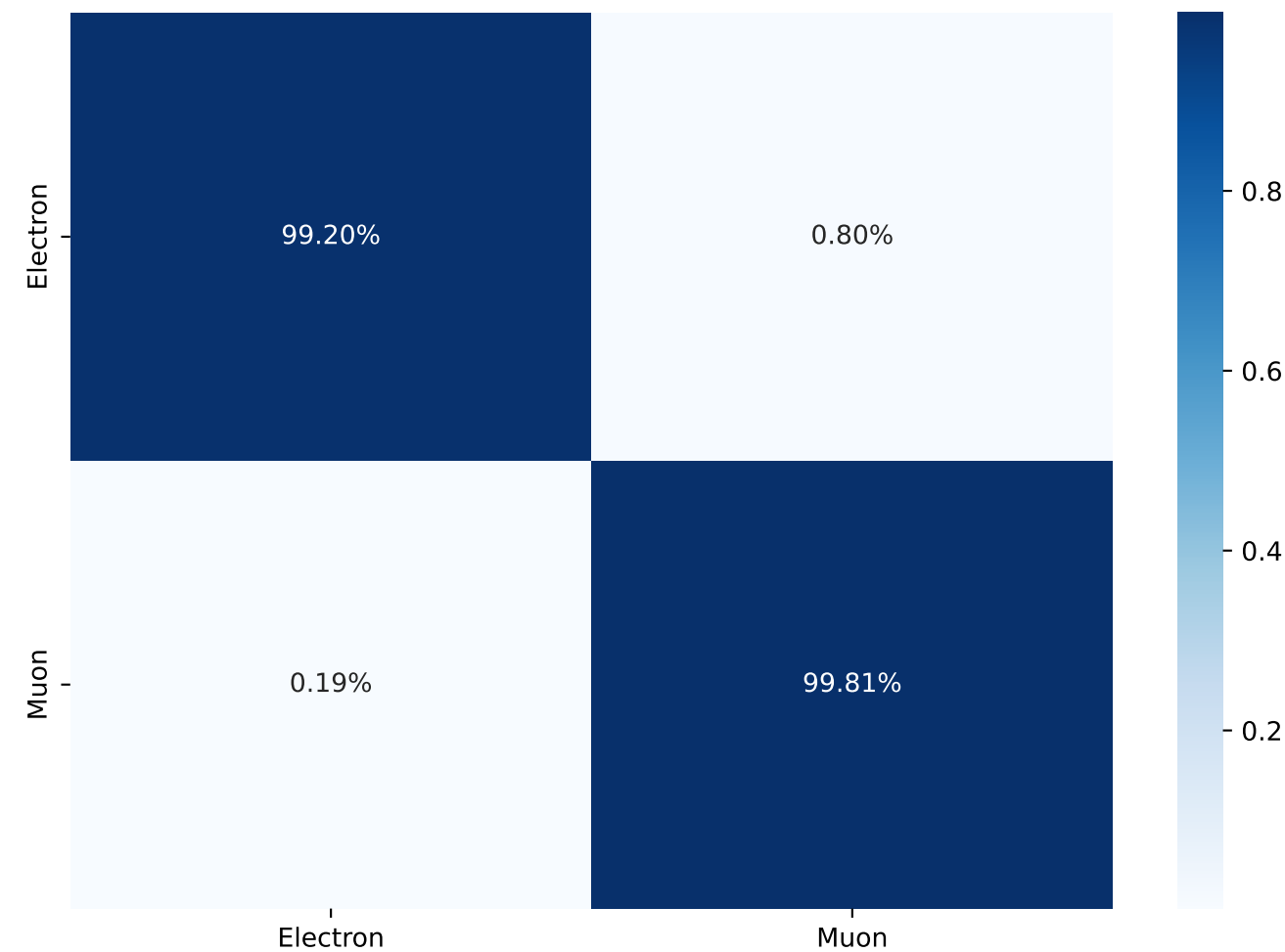


Feature Importance

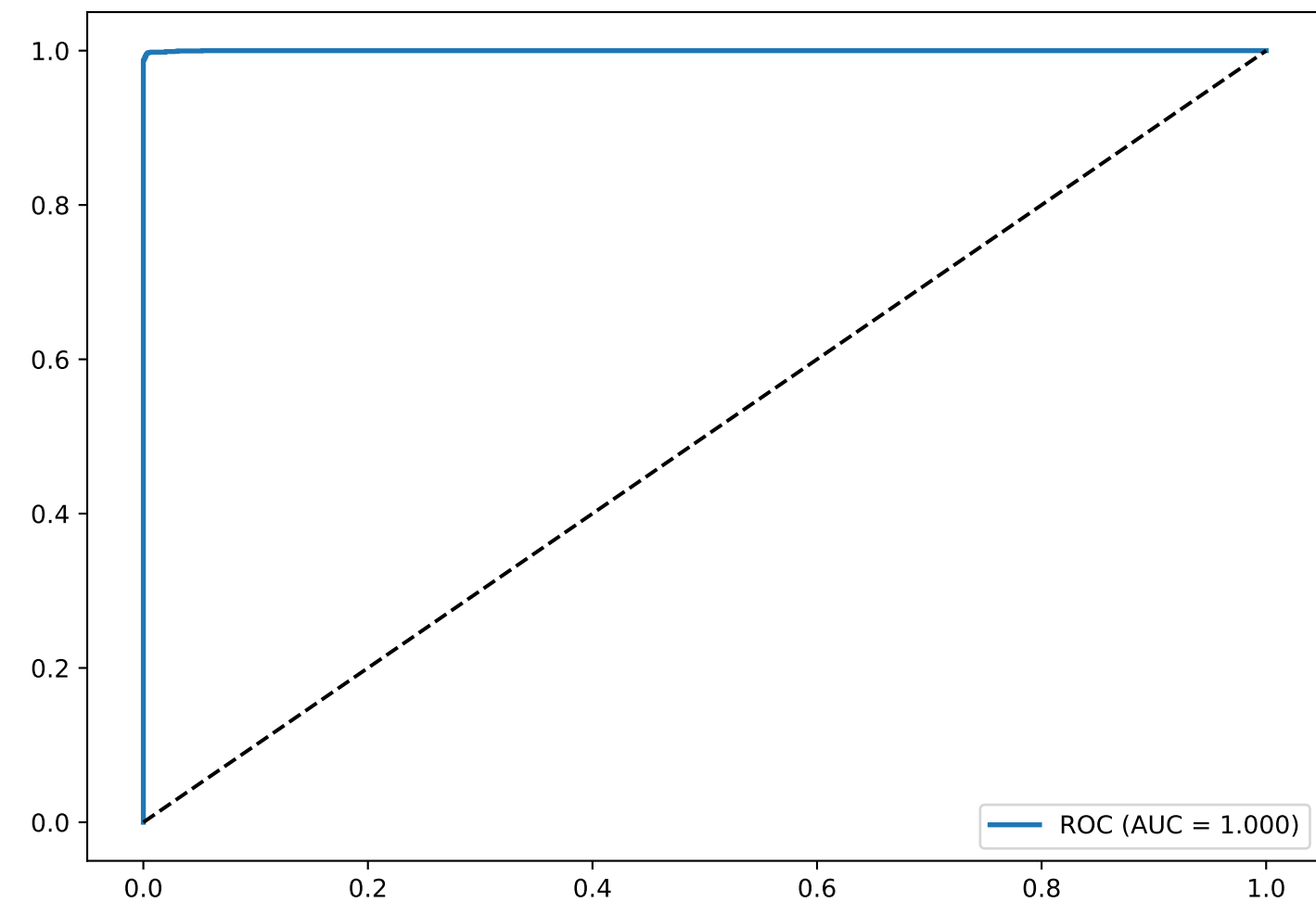


EtaRange = 1.0-1.5 (TEST DATA) - Accuracy: 0.997

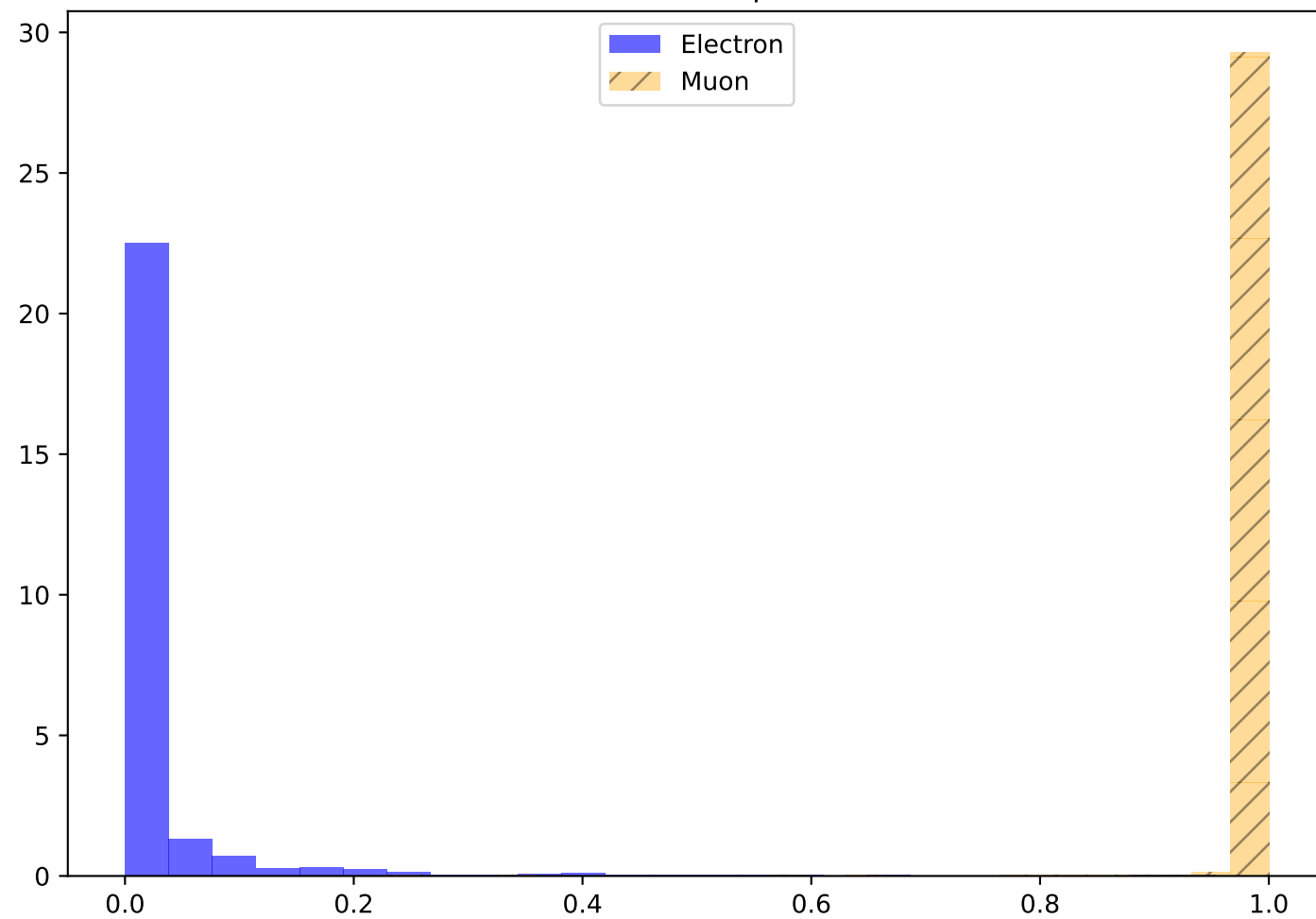
Confusion Matrix



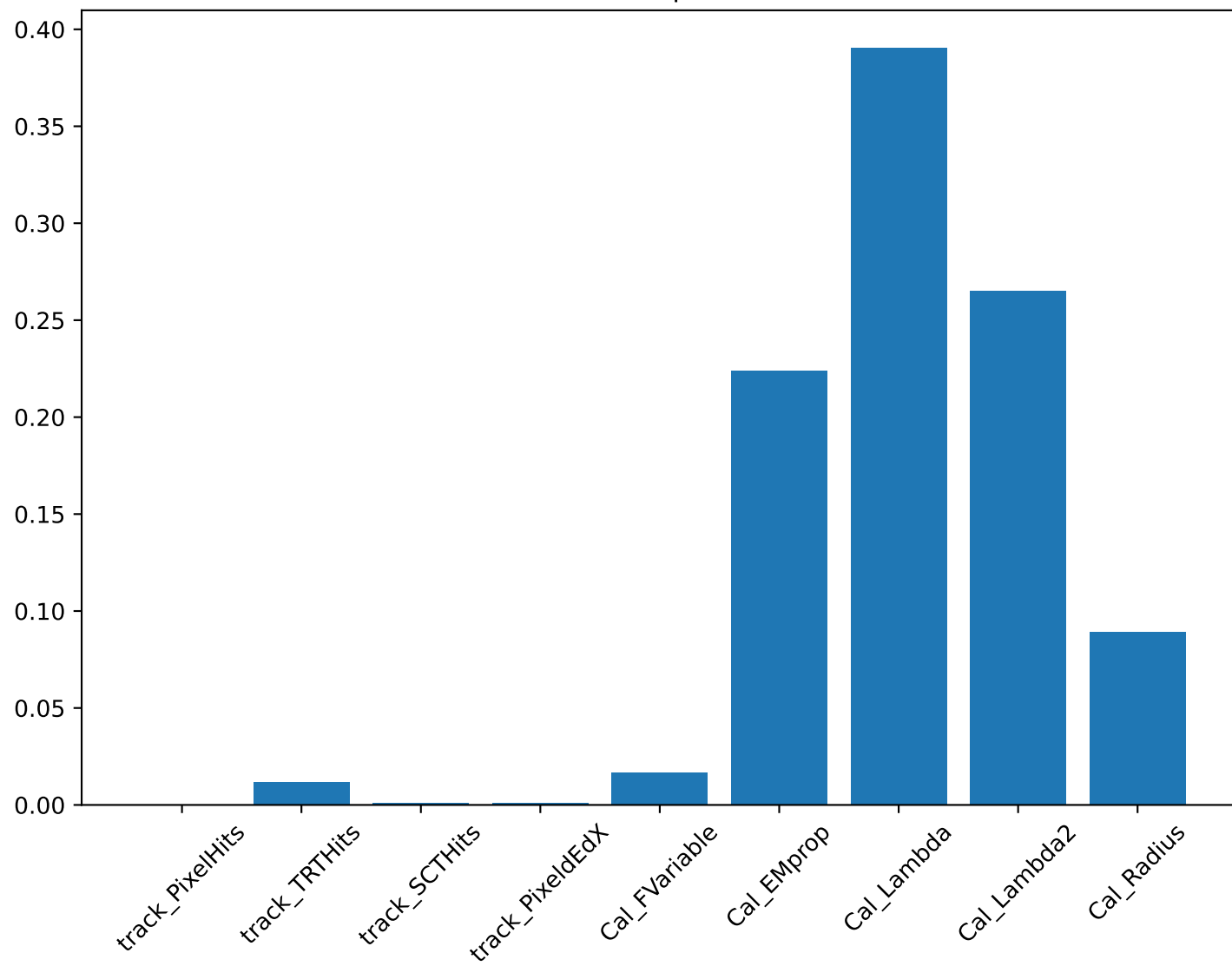
ROC Curve



Classifier Response

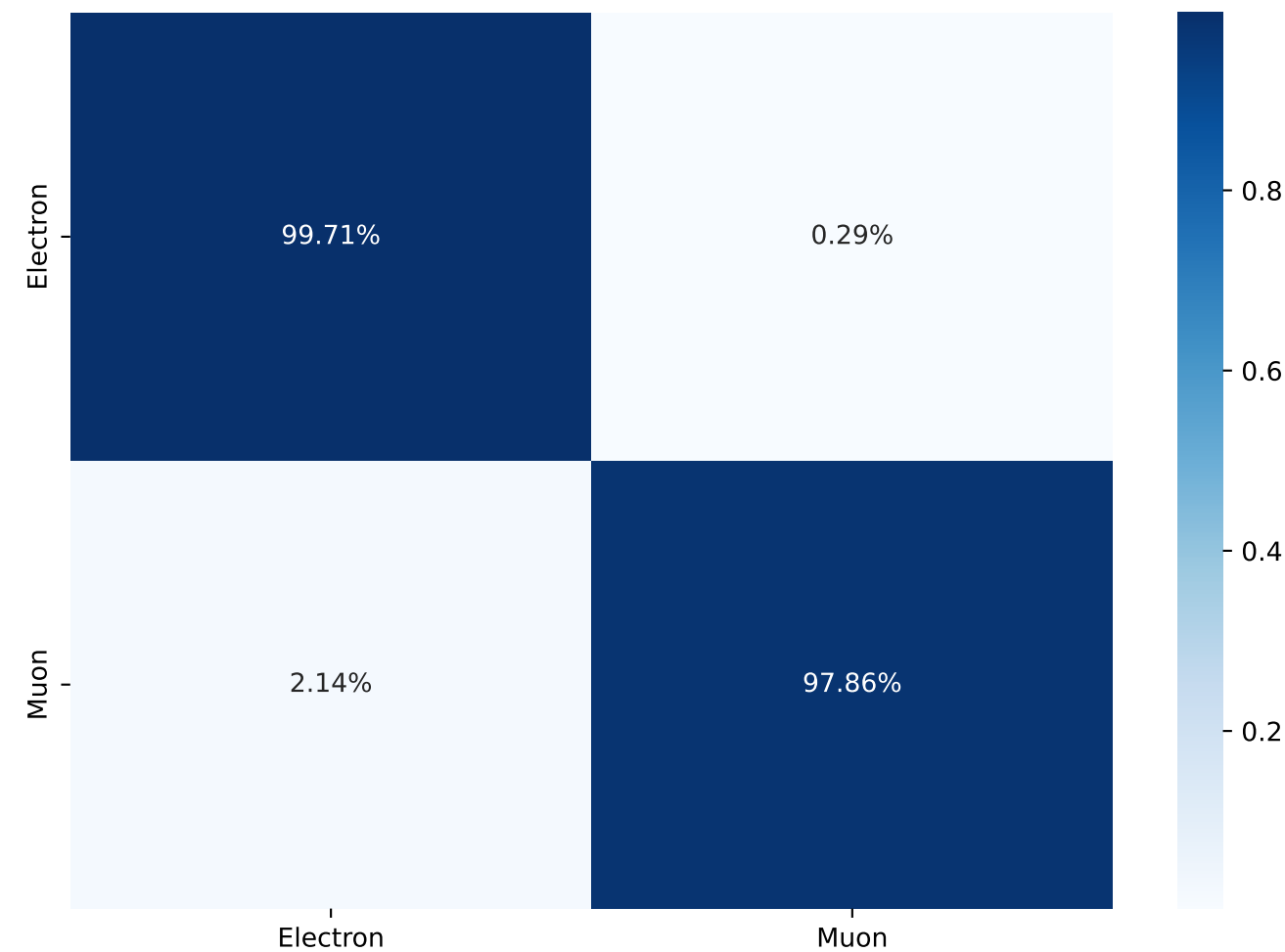


Feature Importance

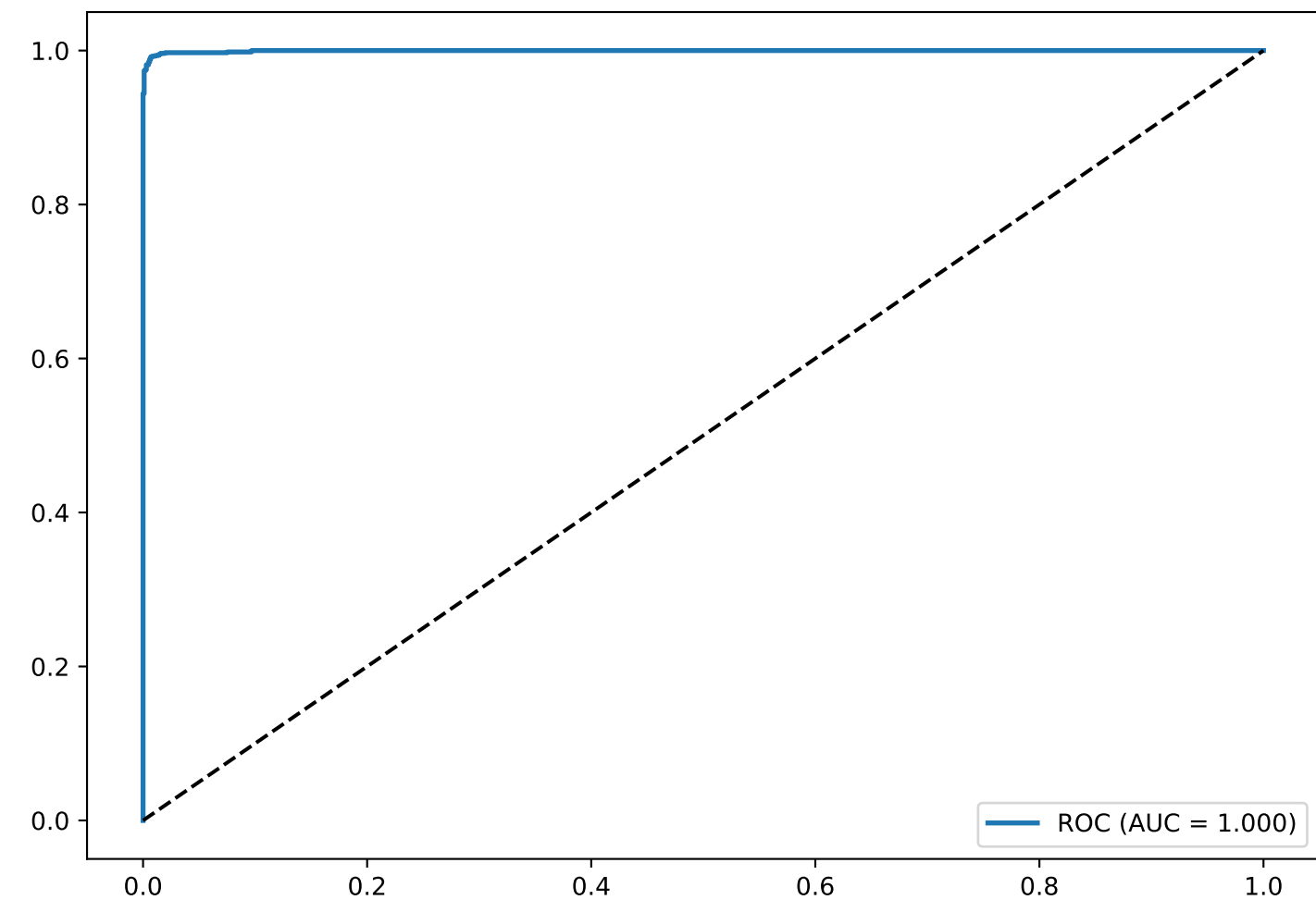


EtaRange = 1.5-2.5 (TEST DATA) - Accuracy: 0.988

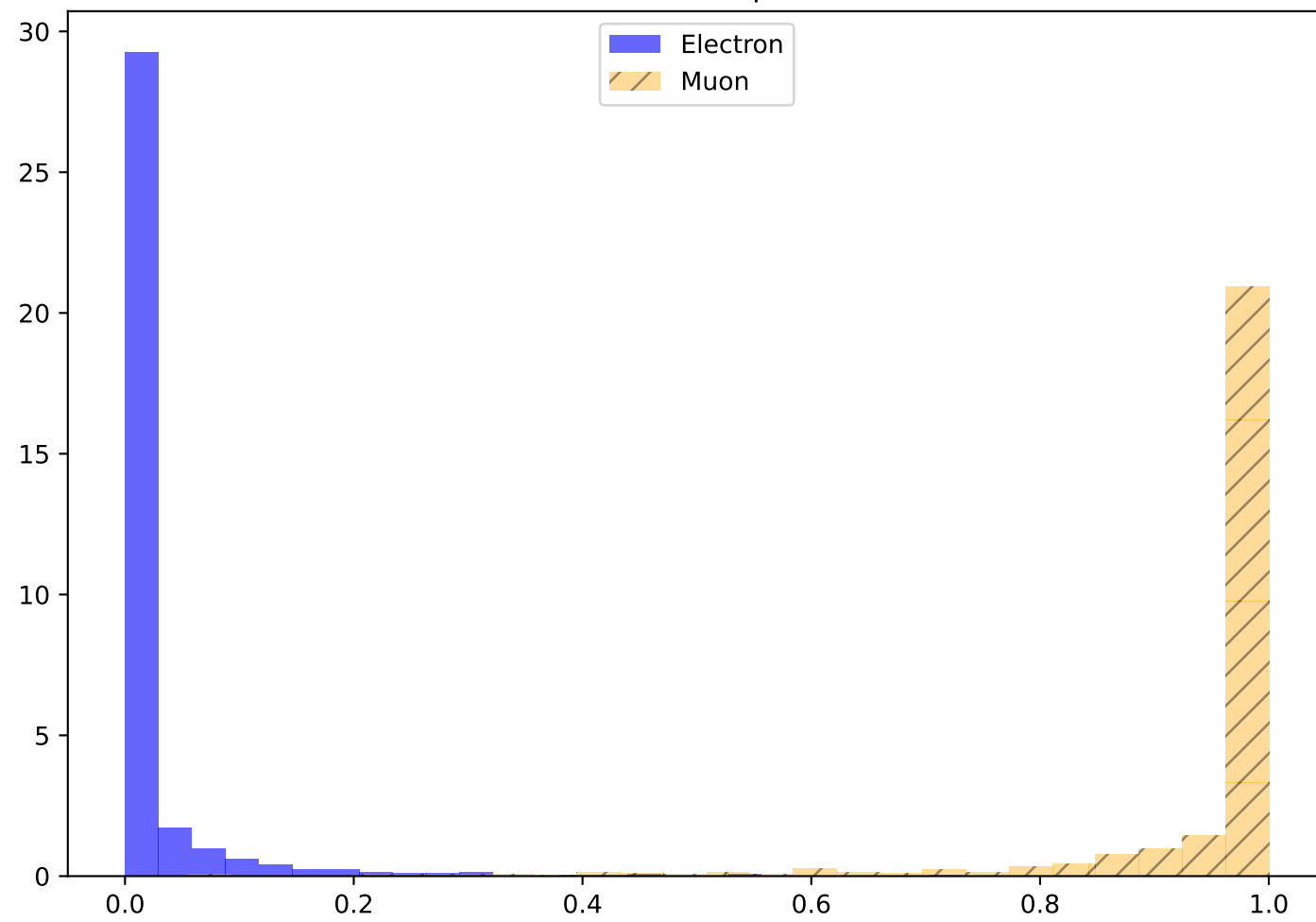
Confusion Matrix



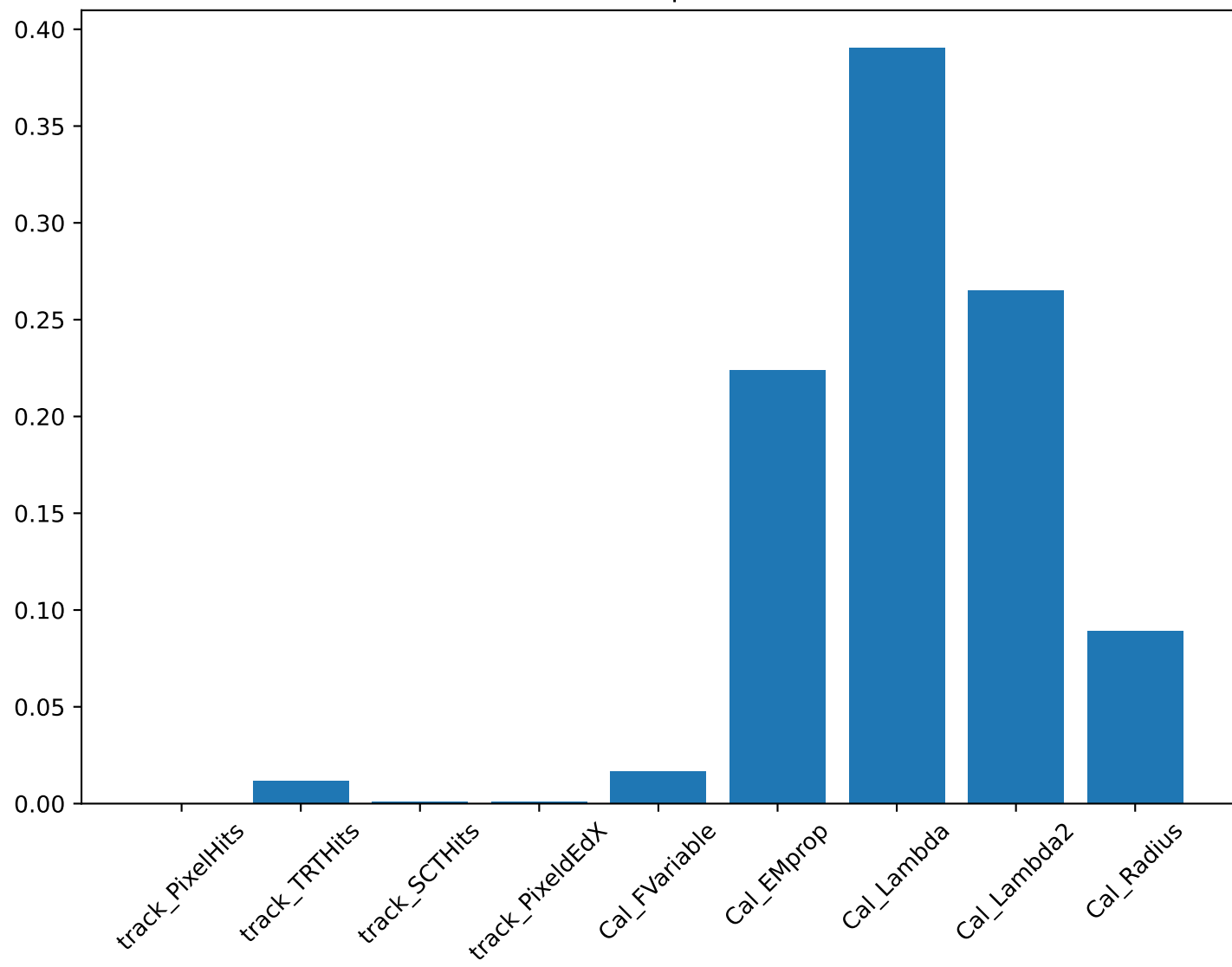
ROC Curve



Classifier Response



Feature Importance



Classification Reports Summary

Trained: Mixed TrueData.root | Test: TrueData.root

EtaRange = FullRange

	precision	recall	f1-score	support
Electron	0.99	1.00	0.99	4813
Muon	1.00	0.99	1.00	8525
accuracy			0.99	13338
macro avg	0.99	0.99	0.99	13338
weighted avg	0.99	0.99	0.99	13338

EtaRange = <1.0

	precision	recall	f1-score	support
Electron	0.99	1.00	0.99	2912
Muon	1.00	0.99	1.00	4877
accuracy			1.00	7789
macro avg	0.99	1.00	1.00	7789
weighted avg	1.00	1.00	1.00	7789

EtaRange = 1.0-1.5

	precision	recall	f1-score	support
Electron	0.99	0.99	0.99	878
Muon	1.00	1.00	1.00	2571
accuracy			1.00	3449
macro avg	1.00	1.00	1.00	3449
weighted avg	1.00	1.00	1.00	3449

EtaRange = 1.5-2.5

	precision	recall	f1-score	support
Electron	0.98	1.00	0.99	1023
Muon	1.00	0.98	0.99	1077
accuracy			0.99	2100
macro avg	0.99	0.99	0.99	2100
weighted avg	0.99	0.99	0.99	2100